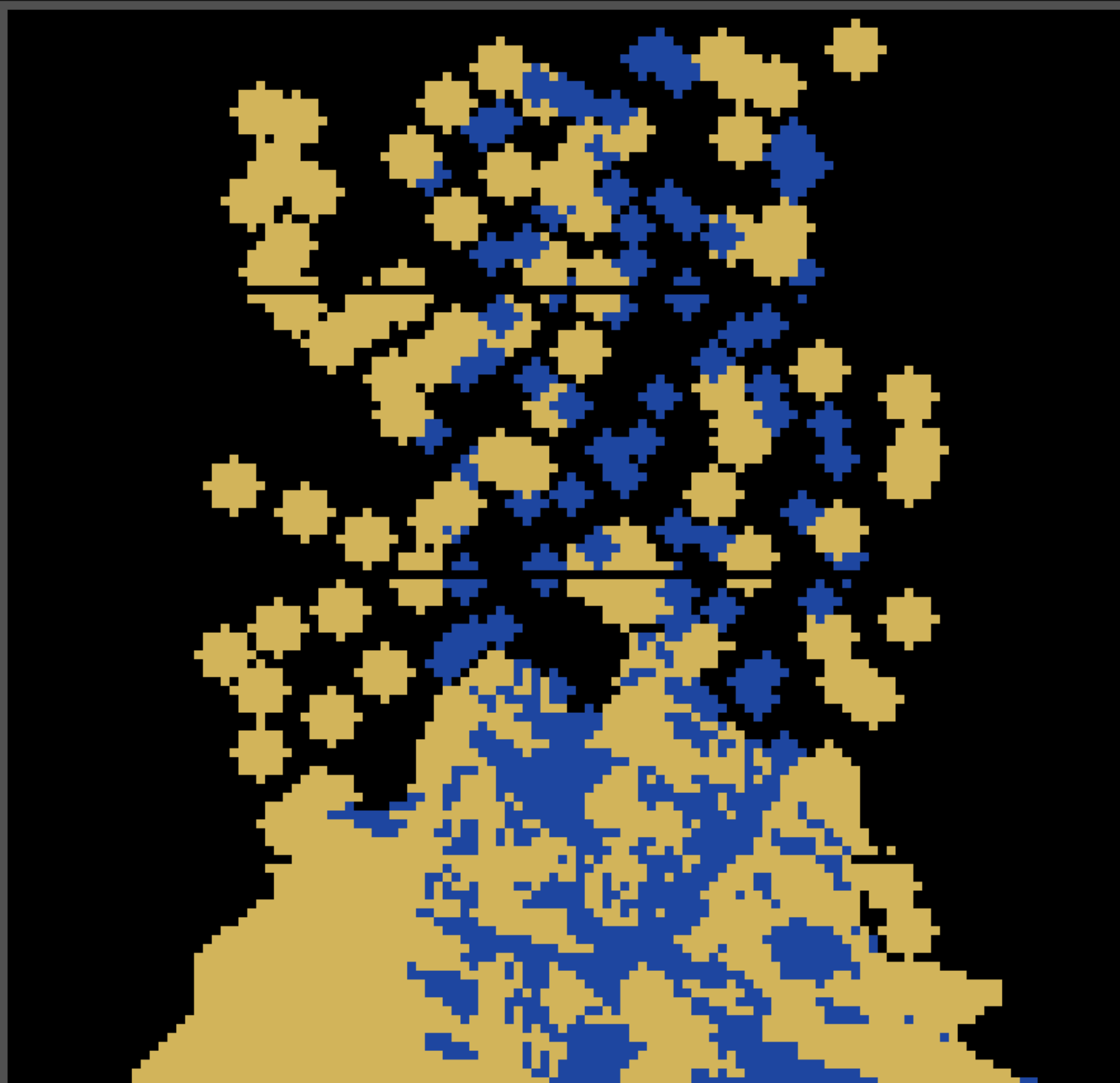


Day 1: a library with no screen

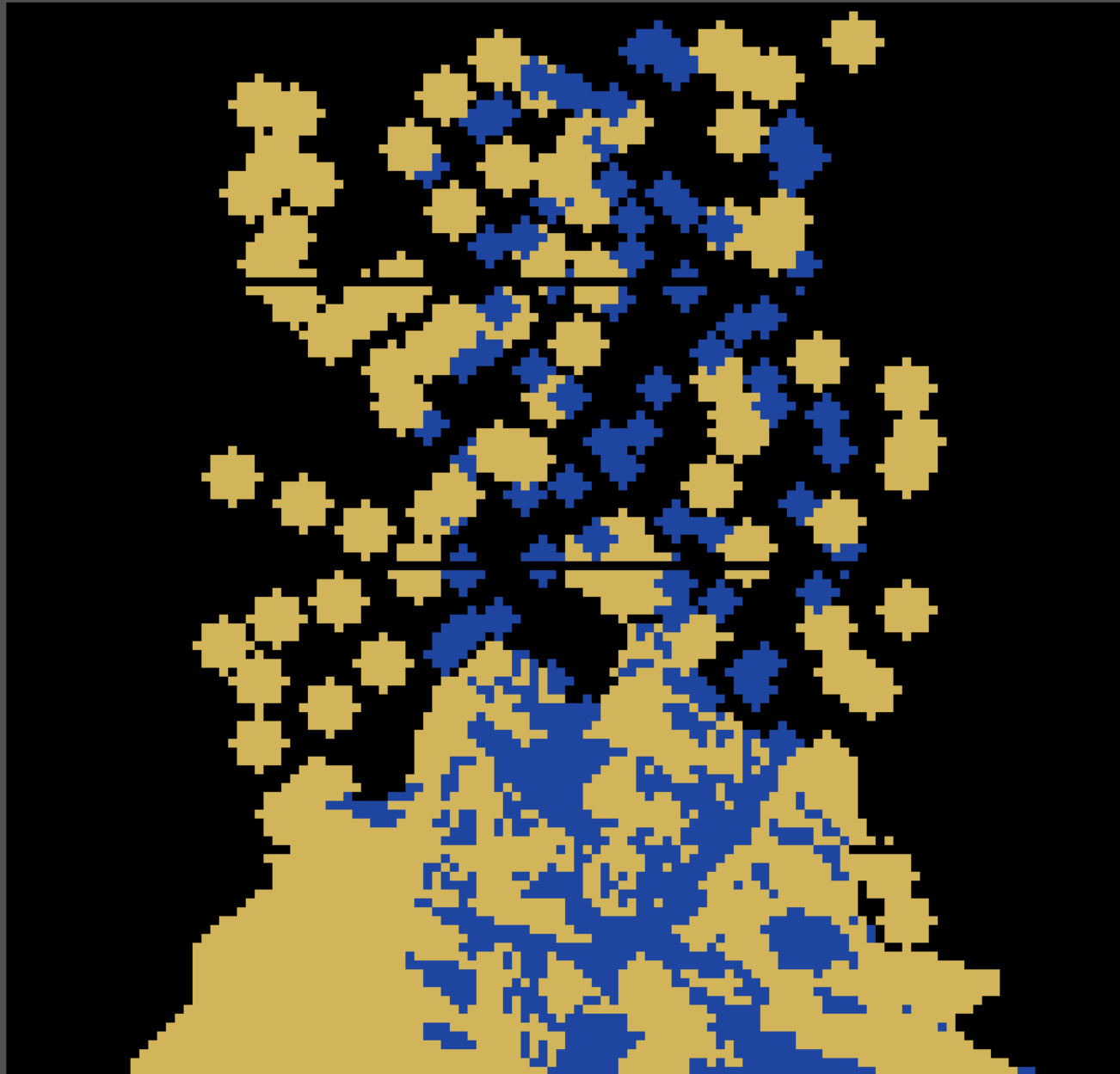


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The first commit

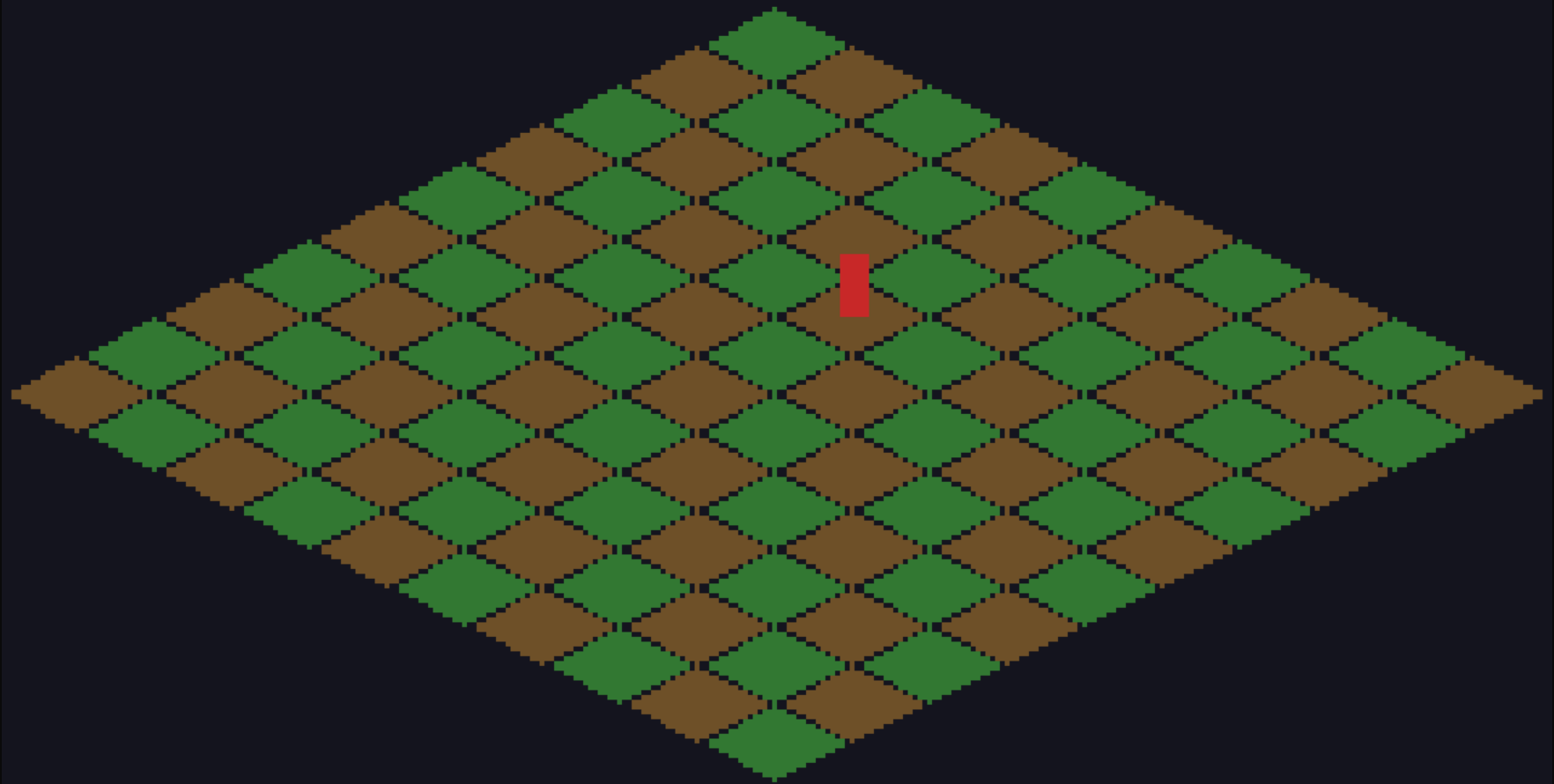
24 files. No game, no engine, no window. Voxelyn started as a headless library for pixels and cellular simulation – and the list of non-goals was written as carefully as the list of goals: no input, no audio, no full physics, no ECS, no UI.

Falling sand



Grid2D packs every cell into 16 bits – material plus flags – chunked, with active and dirty marks. Only active chunks get stepped, so a world at rest costs almost nothing. The demo is that running: sand piling into a slope, water finding its level.

And a green diamond



The other demo: isometric projection, painter order by $x+y$, a red rectangle standing in for a character. Deliberately ugly. What was being tested was whether the draw order closes without seams between tiles – not whether the scene looked good.

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Three rules that held

TypedArrays always. Zero allocation in the hot loop. Deterministic traversal, because a replay has to actually replay. Seven months and a full game later, all three still hold – and that's why the mine floor can be simulated cell by cell on a phone.